Lab 12

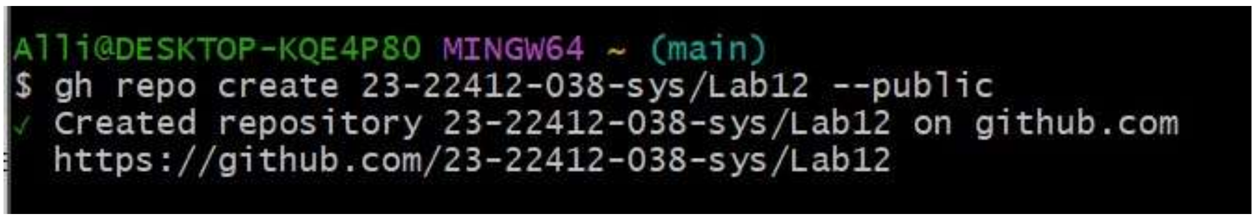
Submitted by: inshal nasir

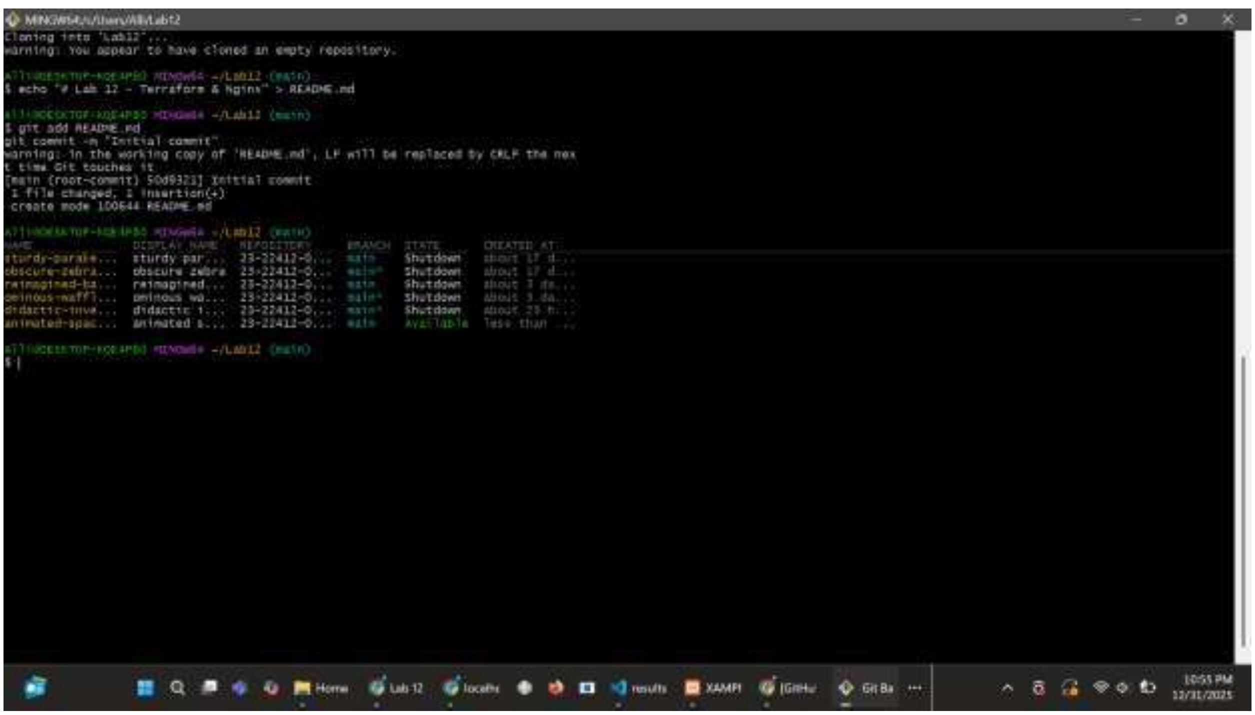
Section : A

Roll No: 38

Submitted to : sir shoaib

Task 0





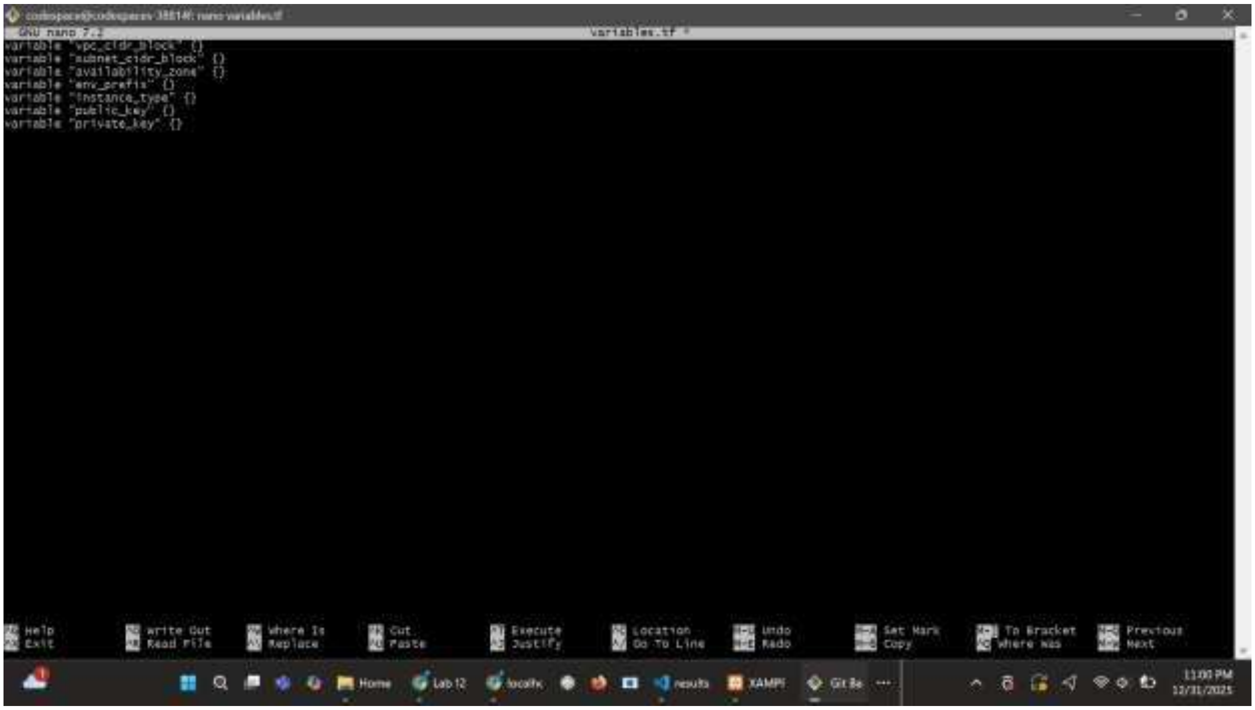
```
codespace@codespaces-38814f: /bin/bash

a11@DESKTOP-KQE4P8O MINGW64 ~/Lab12 (main)
$ echo "# Lab 12 - Terraform & Nginx" > README.md

a11@DESKTOP-KQE4P8O MINGW64 ~/Lab12 (main)
$ git add README.md
git commit -m "Initial commit"
warning: in the working copy of 'README.md', LF will be replaced by CRLF the nex
t time Git touches it
[main (root-commit) 50d8321] Initial commit
 1 file changed, 1 insertion(+)
 create mode 100644 README.md

a11@DESKTOP-KQE4P8O MINGW64 ~/Lab12 (main)
NAME                DISPLAY NAME   REPOSITORY      BRANCH  STATE      CREATED AT
sturdy-parake...    sturdy par...  23-22412-0...   main    Shutdown   about 17 d...
obscure-zebra...    obscure zebra  23-22412-0...   main*   Shutdown   about 17 d...
reimagined-ba...    reimagined...  23-22412-0...   main    Shutdown   about 3 da...
ominous-waffl...    ominous wa...  23-22412-0...   main*   Shutdown   about 3 da...
didactic-inve...    didactic i...  23-22412-0...   main*   Shutdown   about 23 h...
animated-spac...    animated s...  23-22412-0...   main    Available  less than ...

a11@DESKTOP-KQE4P8O MINGW64 ~/Lab12 (main)
$ gh codespace list
NAME                                    DISPLAY NAME           REPOSITORY                            BRANCH  STATE      CREATED AT
sturdy-parakeet-97g95p77v5gp37wv9       sturdy parakeet        23-22412-038-sys/-inshal-nasir-38-    main    Shutdown   about 17 days ago
obscure-zebra-7vj95gvvqx7j3rirg         obscure zebra          23-22412-038-sys/-inshal-nasir-38-    main*   Shutdown   about 17 days ago
reimagined-bassoon-x5gwpj55696q2j9q     reimagined bassoon     23-22412-038-sys/-inshal-nasir-38-    main    Shutdown   about 3 days ago
ominous-waffle-5vvprgrrq55hv9pr         ominous waffle         23-22412-038-sys/-inshal-nasir-38-    main*   Shutdown   about 3 days ago
didactic-invention-x5gwpj556ppg399rw    didactic invention     23-22412-038-sys/Lab11                main*   Shutdown   about 23 hours ago
animated-space-disco-x5gwpj556459hvx9v  animated space disco   23-22412-038-sys/Lab12                main    Available  less than a minute ago

a11@DESKTOP-KQE4P8O MINGW64 ~/Lab12 (main)
$ gh codespace ssh -c animated-space-disco-x5gwpj556459hvx9v
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-1030-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

@23-22412-038-sys → /workspaces/Lab12 (main) $
```

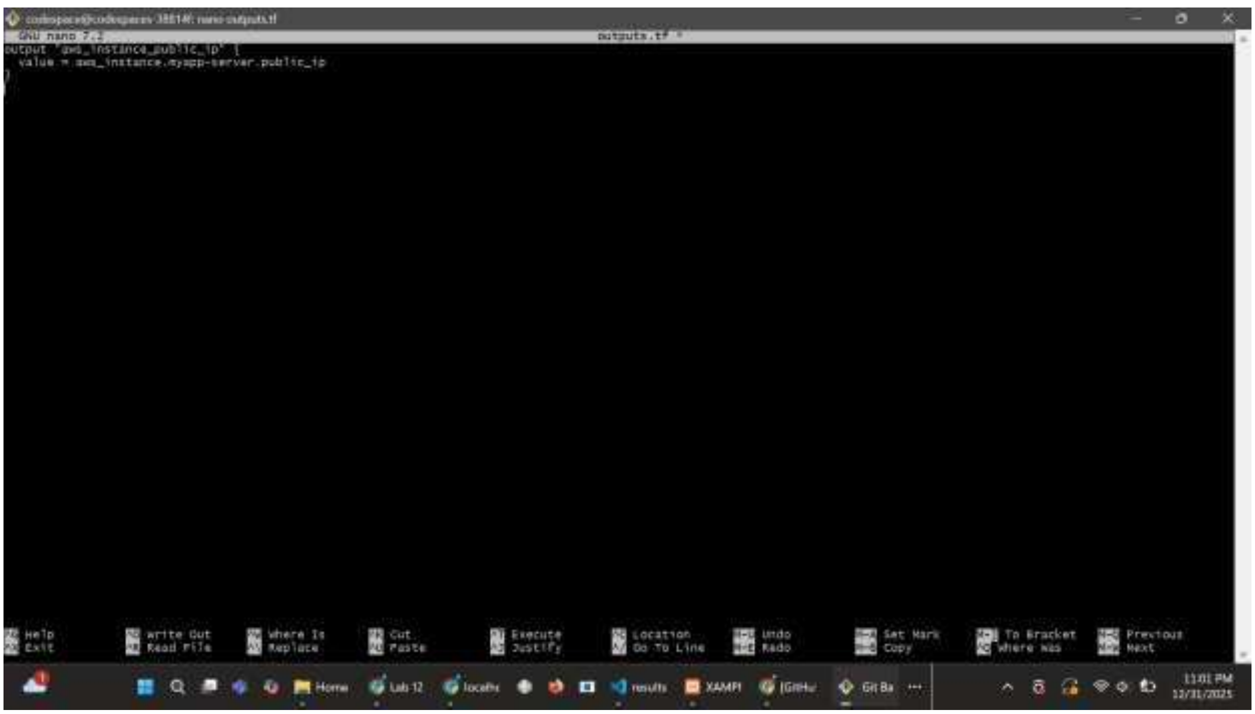
Home | Lab 12 | localhc | results | XAMPP | [GitHu | Git Ba | 10:58 PM 12/31/2025

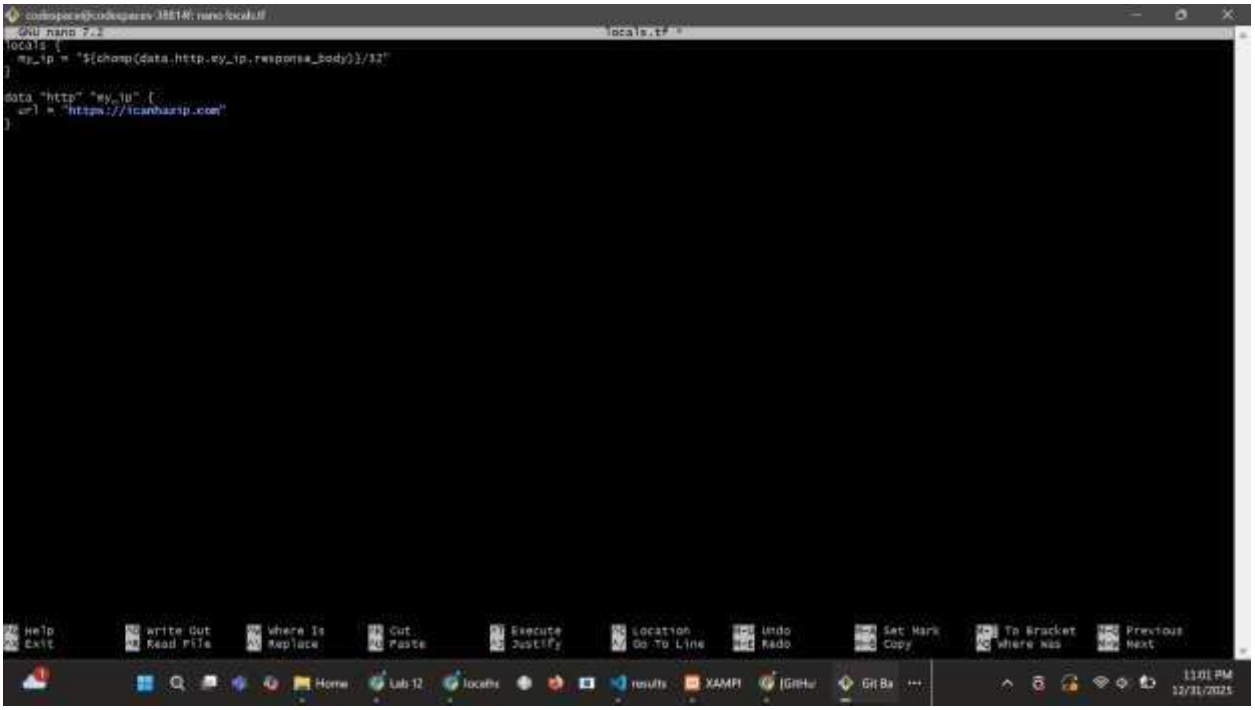
Task 1

```
@23-22412-038-sys → /workspaces/Lab12 (main) $ mkdir -p ~/Lab12
cd ~/Lab12
```

```
@23-22412-038-sys → ~/Lab12 $ touch main.tf variables.tf outputs.tf locals.tf terraform.tfvars entry-script.sh
ls -la
total 12
drwxrwxr-x 2 codespace codespace 4096 Jan  1 06:59 .
drwxr-x--- 1 codespace codespace 4096 Jan  1 06:58 ..
-rw-rw-r-- 1 codespace codespace    0 Jan  1 06:59 entry-script.sh
-rw-rw-r-- 1 codespace codespace    0 Jan  1 06:59 locals.tf
-rw-rw-r-- 1 codespace codespace    0 Jan  1 06:59 main.tf
-rw-rw-r-- 1 codespace codespace    0 Jan  1 06:59 outputs.tf
-rw-rw-r-- 1 codespace codespace    0 Jan  1 06:59 terraform.tfvars
-rw-rw-r-- 1 codespace codespace    0 Jan  1 06:59 variables.tf
```


GNU nano 7.2
variables.tf *
variable "vpc_cidr_block" {}
variable "subnet_cidr_block" {}
variable "availability_zone" {}
variable "env_prefix" {}
variable "instance_type" {}
variable "public_key" {}
variable "private_key" {}
Help
Exit
Write Out
Read File
Where Is
Replace
Cut
Paste
Execute
Justify
Location
Go To Line
Undo
Redo
Set Mark
Copy
To Bracket
Where Was
Previous
Next


GNU nano 7.2
outputs.tf *
output "aws_instance_public_ip" {
value = aws_instance.myapp-server.public_ip
}
Help
Exit
Write Out
Read File
Where Is
Replace
Cut
Paste
Execute
Justify
Location
Go To Line
Undo
Redo
Set Mark
Copy
To Bracket
Where Was
Previous
Next


GNU nano 7.2
locals.tf *
locals {
my_ip = "${chomp(data.http.my_ip.response_body)}/32"
}
data "http" "my_ip" {
url = "https://icanhazip.com"
}
Help
Exit
Write Out
Read File
Where Is
Replace
Cut
Paste
Execute
Justify
Location
Go To Line
Undo
Redo
Set Mark
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To Bracket
Where Was
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Home
Lab 12
results
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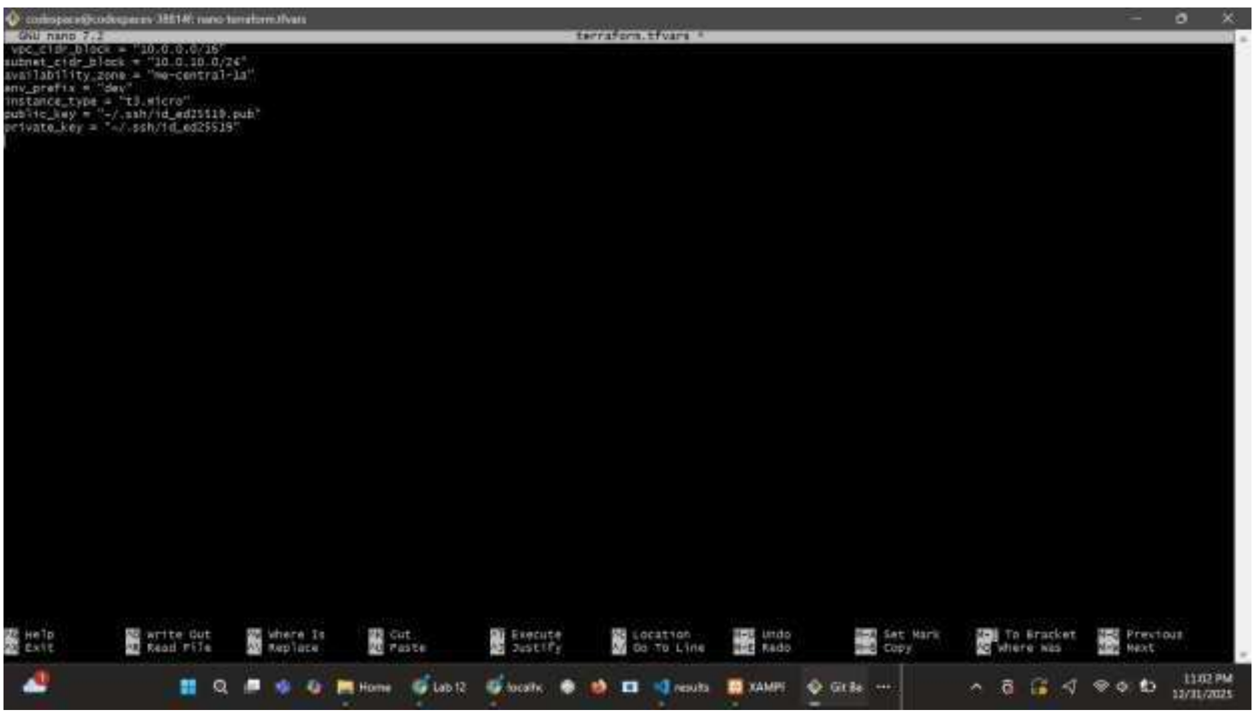
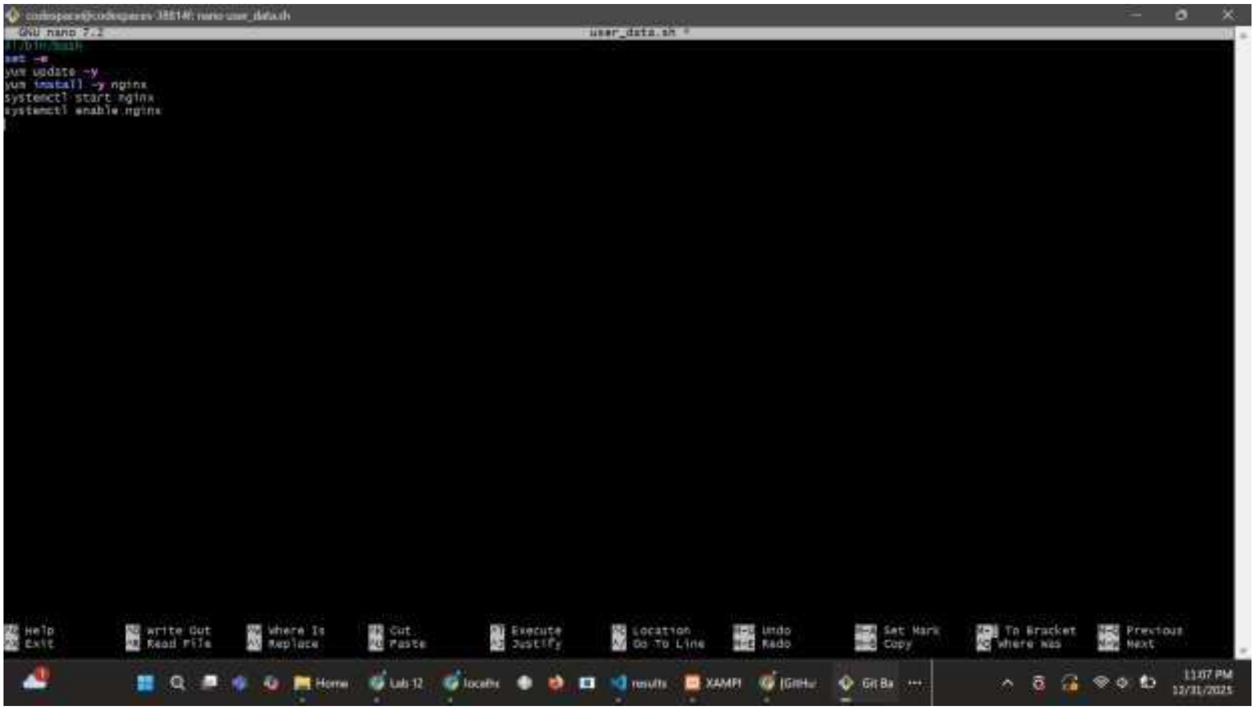
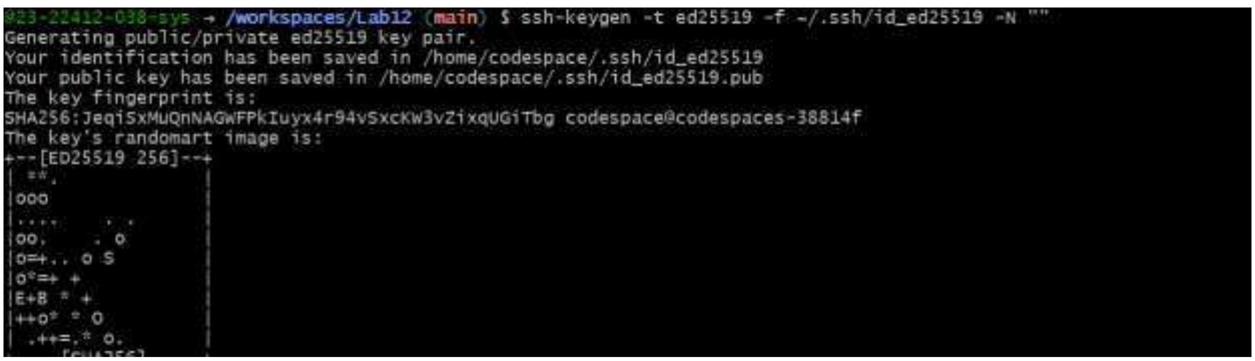
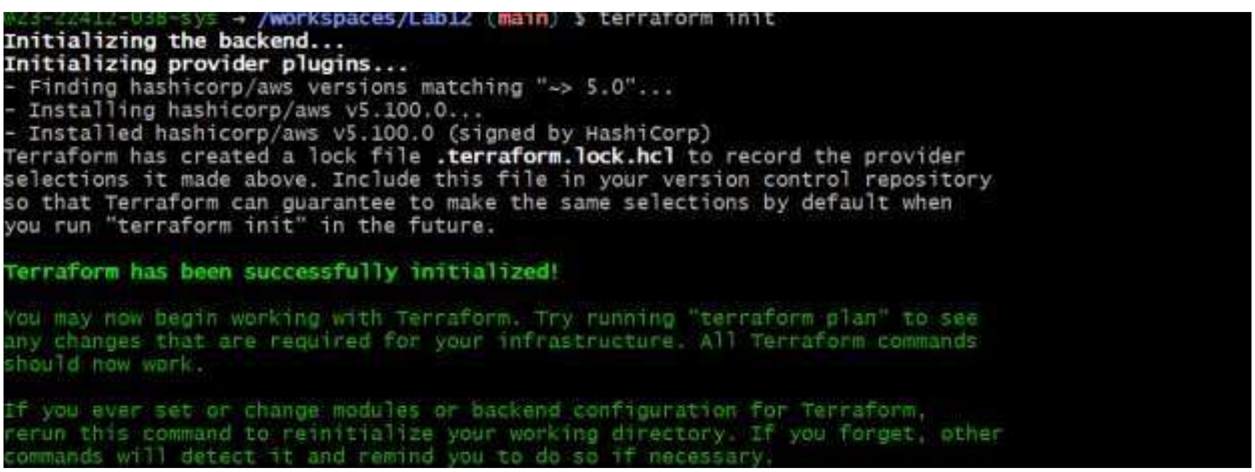
Git Ba
11:01 PM
12/31/2025

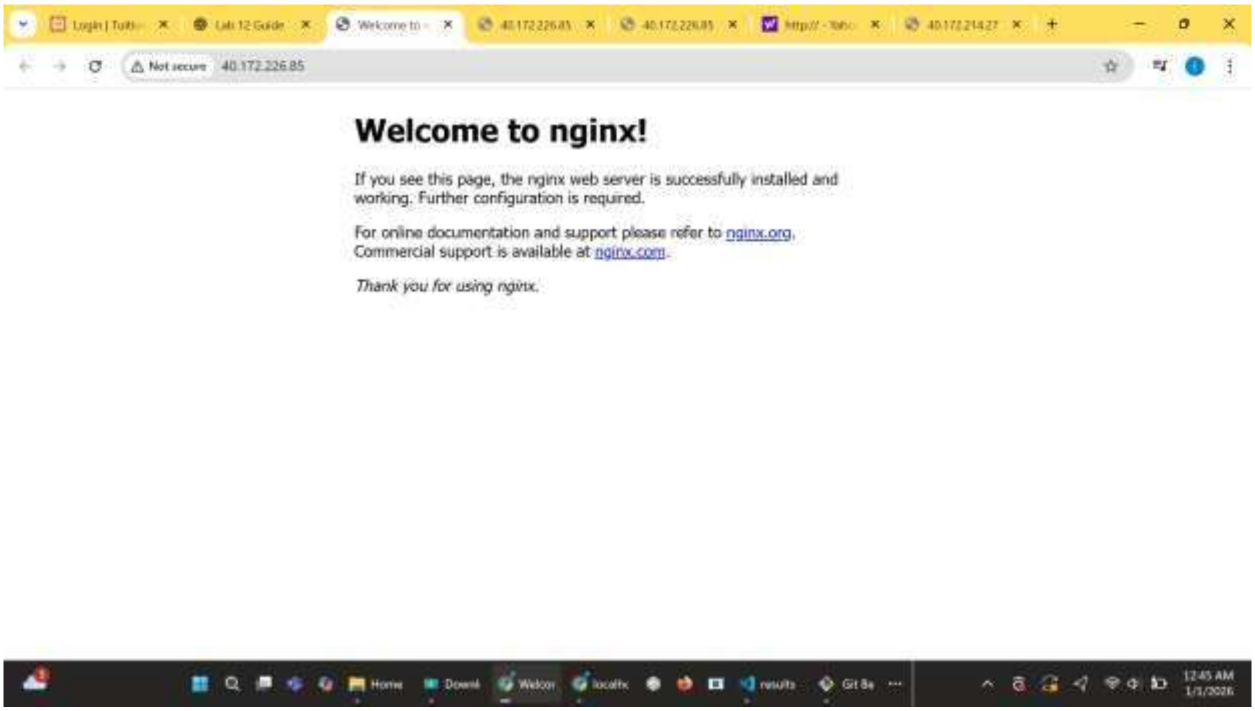

GNU nano 7.2
terraform.tfvars *
vpc_cidr_block = "10.0.0.0/16"
subnet_cidr_block = "10.0.10.0/24"
availability_zone = "me-central-1a"
env_prefix = "dev"
instance_type = "t3.micro"
public_key = "~/.ssh/id_ed25519.pub"
private_key = "~/.ssh/id_ed25519"
Help
Exit
Write Out
Read File
Where Is
Replace
Cut
Paste
Execute
Justify
Location
Go To Line
Undo
Redo
Set Mark
Copy
To Bracket
Where Was
Previous
Next
Home
Lab 12
results
XAMPP
Git Ba
11:02 PM
12/31/2025

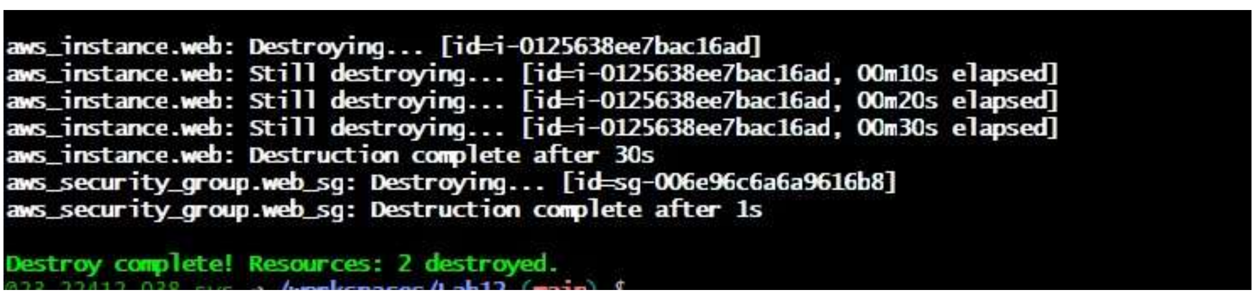

GNU nano 7.2
user_data.sh *
#!/bin/bash
set -e
yum update -y
yum install -y nginx
systemctl start nginx
systemctl enable nginx
Help
Exit
Write Out
Read File
Where Is
Replace
Cut
Paste
Execute
Justify
Location
Go To Line
Undo
Redo
Set Mark
Copy
To Bracket
Where Was
Previous
Next


/workspaces/Lab12 (main) $ ssh-keygen -t ed25519 -f ~/.ssh/id_ed25519 -N ""
Generating public/private ed25519 key pair.
Your identification has been saved in /home/codespace/.ssh/id_ed25519
Your public key has been saved in /home/codespace/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:JeqiSxMuQnNAGWFPkIuyx4r94vSxcKW3vZixqUGiTbg codespace@codespaces-38814f
The key's randomart image is:
+--[ED25519 256]--+


/workspaces/Lab12 (main) $ terraform init
Initializing the backend...
Initializing provider plugins...
- Finding hashicorp/aws versions matching "~> 5.0"...
- Installing hashicorp/aws v5.100.0...
- Installed hashicorp/aws v5.100.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.
Terraform has been successfully initialized!
You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.
If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.

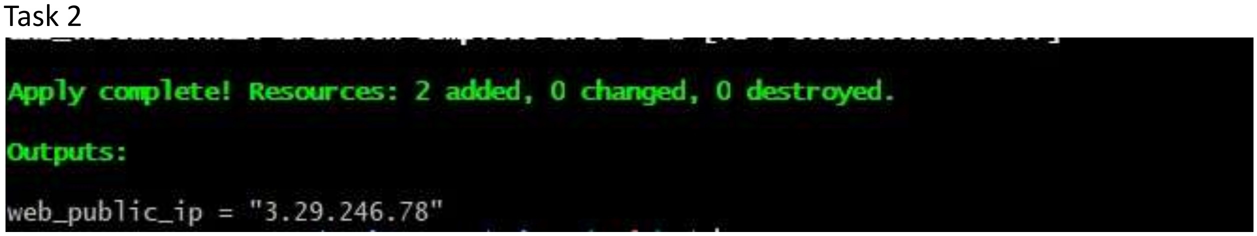
```
Apply complete! Resources: 0 added, 2 changed, 0 destroyed.

Outputs:

web_public_ip = "40.172.226.85"
```



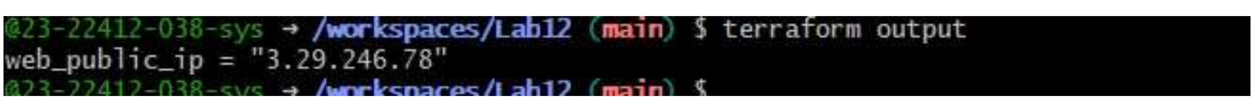


```
aws_instance.web: Destroying... [id=i-0125638ee7bac16ad]
aws_instance.web: Still destroying... [id=i-0125638ee7bac16ad, 00m10s elapsed]
aws_instance.web: Still destroying... [id=i-0125638ee7bac16ad, 00m20s elapsed]
aws_instance.web: Still destroying... [id=i-0125638ee7bac16ad, 00m30s elapsed]
aws_instance.web: Destruction complete after 30s
aws_security_group.web_sg: Destroying... [id=sg-006e96c6a6a9616b8]
aws_security_group.web_sg: Destruction complete after 1s

Destroy complete! Resources: 2 destroyed.
```

Task 2



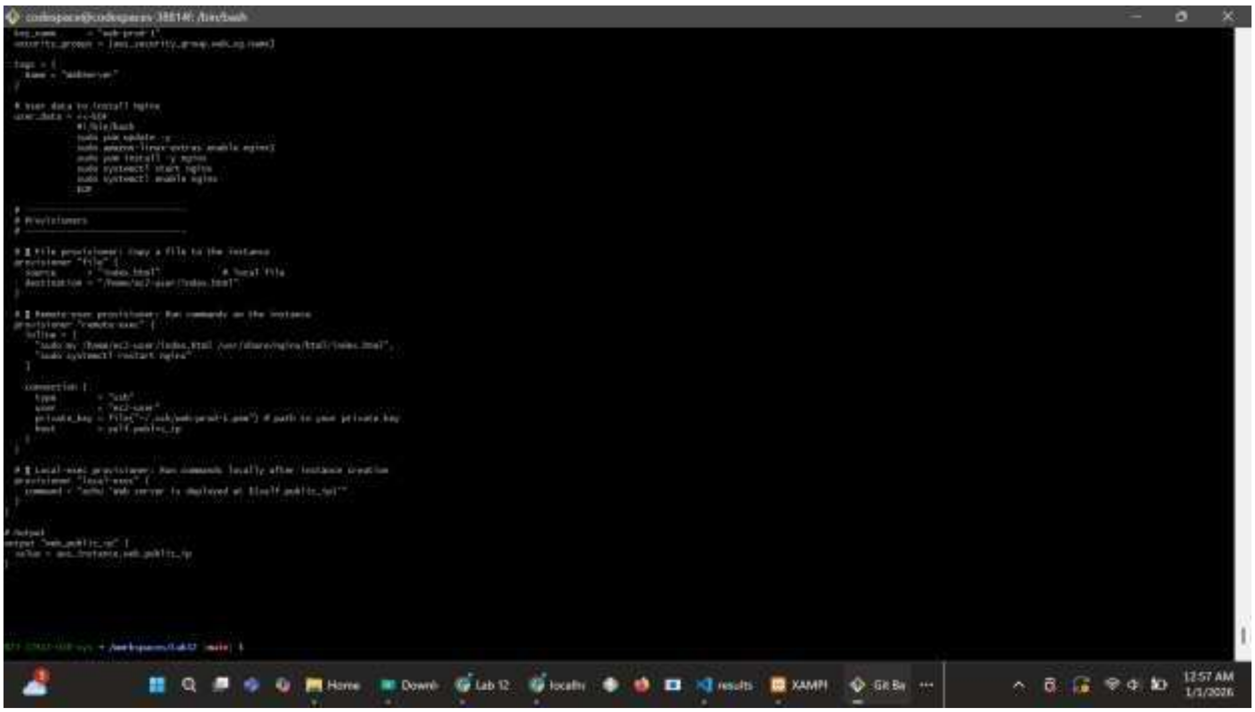
```
Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

Outputs:

web_public_ip = "3.29.246.78"
```

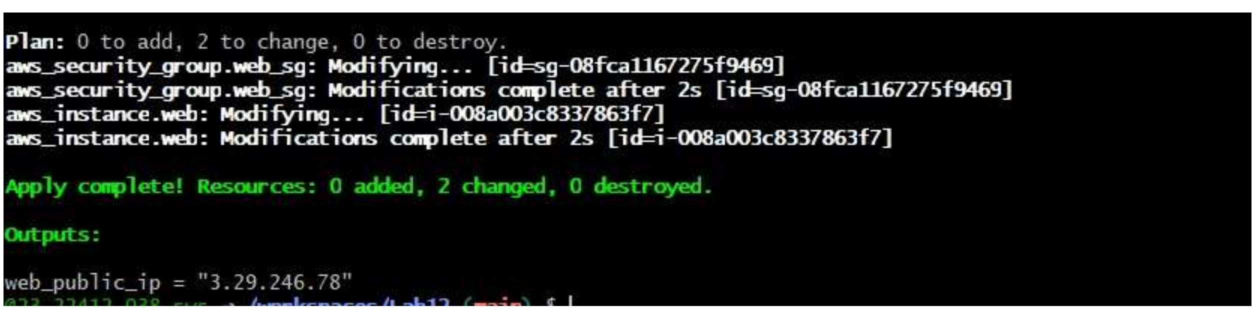


```
@23-22412-038-sys → /workspaces/Lab12 (main) $ terraform output
web_public_ip = "3.29.246.78"
@23-22412-038-sys → /workspaces/Lab12 (main) $
```

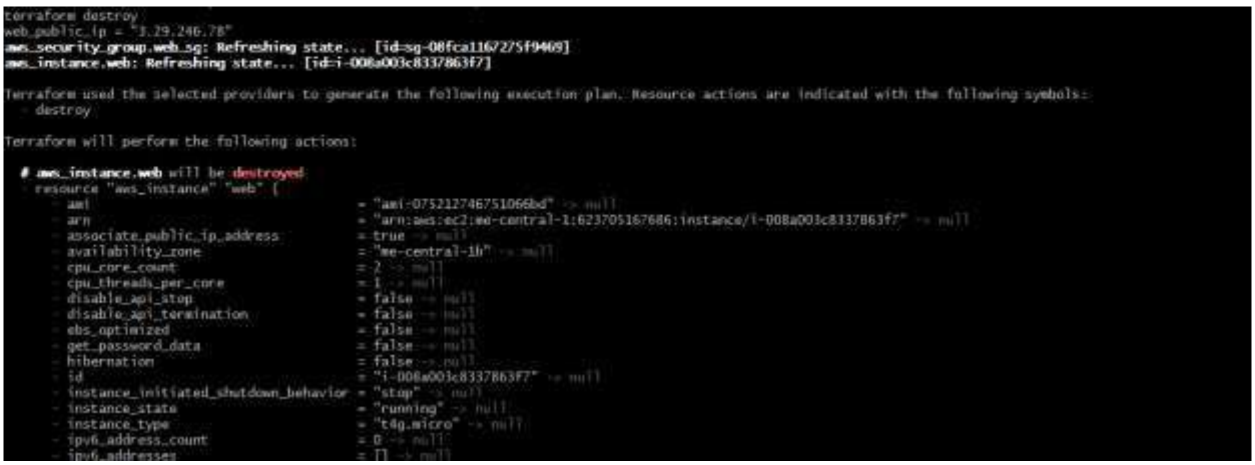

Welcome to nginx!
If you see this page, the nginx web server is successfully installed and working. Further configuration is required.
For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.
Thank you for using nginx.

Task 3

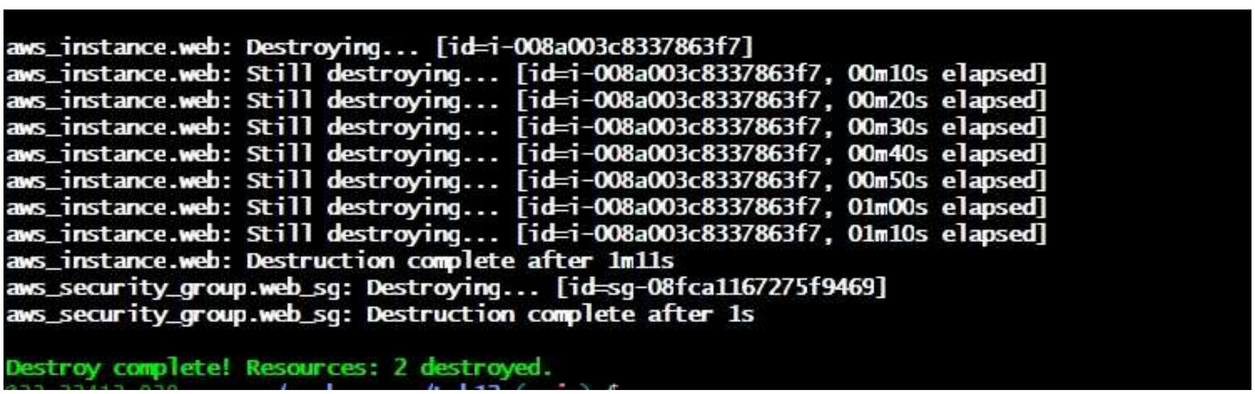



Plan: 0 to add, 2 to change, 0 to destroy.
aws_security_group.web_sg: Modifying... [id=sg-08fca1167275f9469]
aws_security_group.web_sg: Modifications complete after 2s [id=sg-08fca1167275f9469]
aws_instance.web: Modifying... [id=i-008a003c8337863f7]
aws_instance.web: Modifications complete after 2s [id=i-008a003c8337863f7]
Apply complete! Resources: 0 added, 2 changed, 0 destroyed.
Outputs:
web_public_ip = "3.29.246.78"

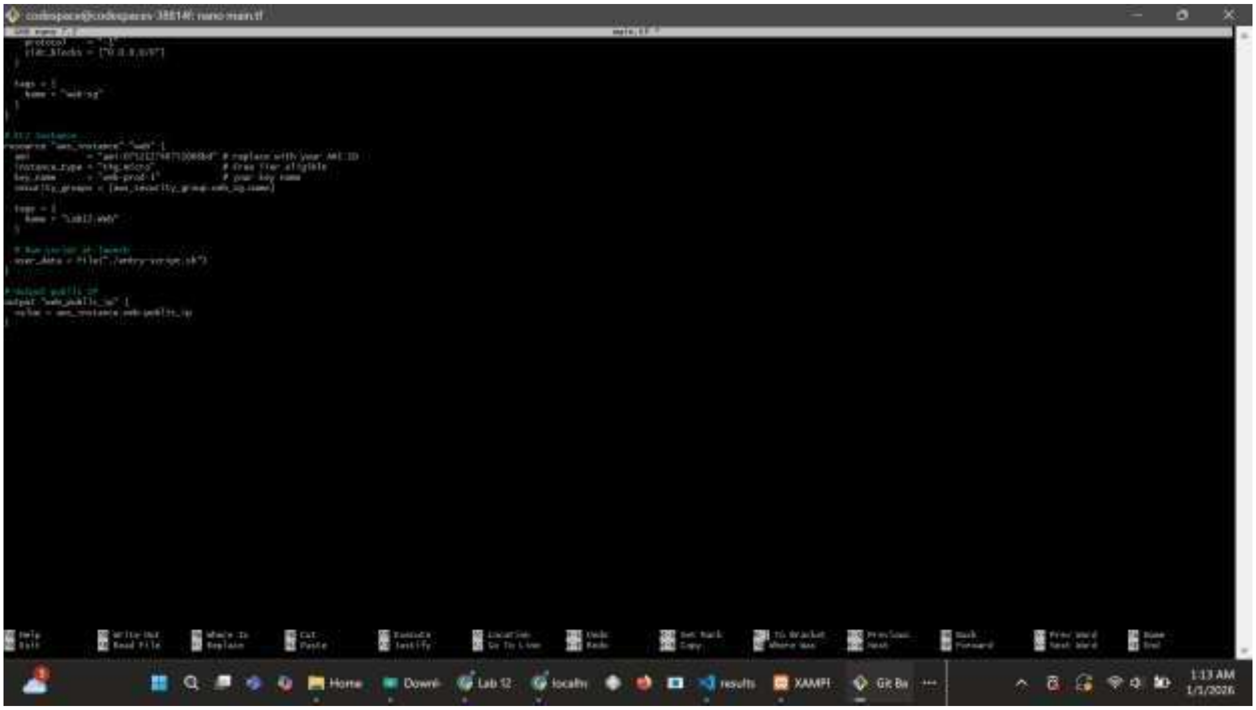
Welcome to nginx!
If you see this page, the nginx web server is successfully installed and working. Further configuration is required.
For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.
Thank you for using nginx.



terraform destroy
web_public_ip = "3.29.246.78"
aws_security_group.web_sg: Refreshing state... [id=sg-08fca1167275f9469]
aws_instance.web: Refreshing state... [id=i-008a003c8337863f7]
Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
- destroy
Terraform will perform the following actions:
# aws_instance.web will be destroyed
- resource "aws_instance" "web" {
- ami = "ami-075212746751066bd" -> null
- arn = "arn:aws:ec2:me-central-1:623705167686:instance/i-008a003c8337863f7" -> null
- associate_public_ip_address = true -> null
- availability_zone = "me-central-1b" -> null
- cpu_core_count = 2 -> null
- cpu_threads_per_core = 1 -> null
- disable_api_stop = false -> null
- disable_api_termination = false -> null
- ebs_optimized = false -> null
- get_password_data = false -> null
- hibernation = false -> null
- id = "i-008a003c8337863f7" -> null
- instance_initiated_shutdown_behavior = "stop" -> null
- instance_state = "running" -> null
- instance_type = "t4g.micro" -> null
- ipv6_address_count = 0 -> null
- ipv6_addresses = [] -> null



aws_instance.web: Destroying... [id=i-008a003c8337863f7]
aws_instance.web: Still destroying... [id=i-008a003c8337863f7, 00m10s elapsed]
aws_instance.web: Still destroying... [id=i-008a003c8337863f7, 00m20s elapsed]
aws_instance.web: Still destroying... [id=i-008a003c8337863f7, 00m30s elapsed]
aws_instance.web: Still destroying... [id=i-008a003c8337863f7, 00m40s elapsed]
aws_instance.web: Still destroying... [id=i-008a003c8337863f7, 00m50s elapsed]
aws_instance.web: Still destroying... [id=i-008a003c8337863f7, 01m00s elapsed]
aws_instance.web: Still destroying... [id=i-008a003c8337863f7, 01m10s elapsed]
aws_instance.web: Destruction complete after 1m11s
aws_security_group.web_sg: Destroying... [id=sg-08fca1167275f9469]
aws_security_group.web_sg: Destruction complete after 1s
Destroy complete! Resources: 2 destroyed.



Task 4

```
variable "vpc_id" {}
variable "subnet_cidr_block" {}
variable "availability_zone" {}
variable "env_prefix" {}
variable "default_route_table_id" {}
```

```
resource "aws_subnet" "myapp_subnet_1" {
  vpc_id                  = var.vpc_id
  cidr_block              = var.subnet_cidr_block
  availability_zone       = var.availability_zone
  map_public_ip_on_launch = true
  tags = {
     Name = "${var.env_prefix}-subnet-1"
  }
}

resource "aws_internet_gateway" "myapp_igw" {
  vpc_id = var.vpc_id
  tags = {
     Name = "${var.env_prefix}-igw"
  }
}

resource "aws_default_route_table" "main_rt" {
  default_route_table_id = var.default_route_table_id

  route {
    cidr_block = "0.0.0.0/0"
    gateway_id = aws_internet_gateway.myapp_igw.id
  }
  tags = {
     Name = "${var.env_prefix}-rt"
  }
}
```

```
output "subnet" {
  value = aws_subnet.myapp_subnet_1
}
```

```
module "myapp-subnet" {
  source                  = "./modules/subnet"
  vpc_id                  = aws_vpc.myapp_vpc.id
  subnet_cidr_block       = var.subnet_cidr_block
  availability_zone       = var.availability_zone
  env_prefix              = var.env_prefix
  default_route_table_id = aws_vpc.myapp_vpc.default_route_table_id
}
```

```
Initializing the backend...
Initializing modules...
- myapp-subnet in modules/subnet
Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Reusing previous version of hashicorp/http from the dependency lock file
- Using previously-installed hashicorp/aws v6.28.0
- Using previously-installed hashicorp/http v3.5.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
```

```
module.myapp-subnet.aws_subnet.myapp_subnet_1: Creating...
aws_default_security_group.default_sg: Creating...
module.myapp-subnet.aws_internet_gateway.myapp_igw: Creation complete after 1s [id=igw-0a7894f04dfd9a537]
module.myapp-subnet.aws_default_route_table.main_rt: Creating...
module.myapp-subnet.aws_default_route_table.main_rt: Creation complete after 1s [id=rtb-05b894e5b39dd25a8]
aws_default_security_group.default_sg: Creation complete after 4s [id=sg-0f8b4a66badedc8ca]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Still creating... [00m10s elapsed]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Creation complete after 11s [id=subnet-0e29387ac22e28b6e]
aws_instance.myapp-server: Creating...
aws_instance.myapp-server: Still creating... [00m10s elapsed]
aws_instance.myapp-server: Creation complete after 14s [id=i-082355d8b3da69cc0]

Apply complete! Resources: 7 added, 0 changed, 0 destroyed.

Outputs:

aws_instance_public_ip = "51.112.167.169"
```

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Task 5

```
$ mkdir -p modules/webserver
$ touch modules/webserver/main.tf modules/webserver/variables.tf modules/webserver/outputs.tf
$ ls -R modules/webserver

$
```

modules > webserver > variables.tf

```
variable "availability_zone" {}
variable "public_key" {}
variable "my_ip" {}
variable "vpc_id" {}
variable "subnet_id" {}
variable "script_path" {}
variable "instance_suffix" {}
```

```
resource "aws_security_group" "web_sg" {
  vpc_id      = var.vpc_id
  name        = "${var.env_prefix}-web-sg-${var.instance_suffix}"
  description = "Security group for web server allowing HTTP, HTTPS and SSH"

  ingress {
    from_port   = 22
    to_port     = 22
    protocol    = "tcp"
    cidr_blocks = [var.my_ip]
  }
  ingress {
    from_port   = 443
    to_port     = 443
    protocol    = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }
  ingress {
    from_port   = 80
    to_port     = 80
    protocol    = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }
  egress {
    from_port   = 0
    to_port     = 0
    protocol    = "-1"
    cidr_blocks = ["0.0.0.0/0"]
  }
  tags = { Name = "${var.env_prefix}-web-sg-${var.instance_suffix}" }
}

resource "aws_key_pair" "ssh-key" {
```

```
output "aws_instance" {
  value = aws_instance.myapp-server
}
```

```
output "webserver_public_ip" {
  value = module.myapp-webserver.aws_instance.public_ip
}
```

```
provider "aws" {
  region = "me-central-1"
}

resource "aws_vpc" "myapp_vpc" {
  cidr_block = var.vpc_cidr_block
  tags = { Name = "${var.env_prefix}-vpc" }
}

module "myapp-subnet" {
  source                 = "./modules/subnet"
  vpc_id                 = aws_vpc.myapp_vpc.id
  subnet_cidr_block      = var.subnet_cidr_block
  availability_zone      = var.availability_zone
  env_prefix             = var.env_prefix
  default_route_table_id = aws_vpc.myapp_vpc.default_route_table_id
}

module "myapp-webserver" {
  source            = "./modules/webserver"
  env_prefix        = var.env_prefix
  instance_type     = var.instance_type
  availability_zone = var.availability_zone
  public_key        = var.public_key
  my_ip             = local.my_ip
  vpc_id            = aws_vpc.myapp_vpc.id
  subnet_id         = module.myapp-subnet.subnet.id
  script_path       = "./entry-script.sh"
  instance_suffix   = "0"
}
```

```
Initializing the backend...
Initializing modules...
- myapp-webserver in modules/webserver
Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Reusing previous version of hashicorp/http from the dependency lock file
- Using previously-installed hashicorp/aws v6.28.0
- Using previously-installed hashicorp/http v3.5.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
```

```
Apply complete! Resources: 3 added, 0 changed, 3 destroyed.

Outputs:

webserver_public_ip = "51.112.231.203"
```

```
webserver_public_ip = "51.112.231.203"
```

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Task 6

```
entry-script.sh
#!/bin/bash
set -e

# Install and Start Nginx
yum update -y
yum install -y nginx
systemctl start nginx
systemctl enable nginx

# Create directories for SSL certificates
mkdir -p /etc/ssl/private
mkdir -p /etc/ssl/certs

# Get IMDSv2 token for security
TOKEN=$(curl -s -X PUT "http://169.254.169.254/latest/api/token" -H "X-aws-ec2-metadata-token-ttl-seconds: 21600")

# Get current public IP dynamically
PUBLIC_IP=$(curl -s -H "X-aws-ec2-metadata-token: $TOKEN" http://169.254.169.254/latest/meta-data/public-ipv4)

# Generate self-signed certificate with the Public IP as the Common Name (CN)
openssl req -x509 -nodes -days 365 -newkey rsa:2048 \
  -keyout /etc/ssl/private/selfsigned.key \
  -out /etc/ssl/certs/selfsigned.crt \
  -subj "/CN=$PUBLIC_IP" \
  -addext "subjectAltName=IP:$PUBLIC_IP" \
  -addext "basicConstraints=CA:FALSE" \
  -addext "keyUsage=digitalSignature,keyEncipherment" \
  -addext "extendedKeyUsage=serverAuth"

# Configure Nginx for SSL and Redirects
cat <<EOF > /etc/nginx/nginx.conf
user nginx;
worker_processes auto;
```

```
Apply complete! Resources: 7 added, 0 changed, 0 destroyed.

Outputs:

webserver_public_ip = "3.28.136.167"
```

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Task 7

```
yum update -y
yum install httpd -y
systemctl start httpd
systemctl enable httpd

echo "<h1>Welcome to My Web Server</h1>" > /var/www/html/index.html
hostnamectl set-hostname myapp-webserver
echo "<h2>Hostname:  $(hostname)</h2>" >> /var/www/html/index.html

TOKEN=$(curl -s -X PUT "http://169.254.169.254/latest/api/token" -H "X-aws-ec2-metadata-token-ttl-seconds: 21600")

echo "<h2>Private IP: $(curl -s -H "X-aws-ec2-metadata-token: $TOKEN" http://169.254.169.254/latest/meta-data/local-ipv4)</h2>" >> /var/www/html/index.
echo "<h2>Public IP: $(curl -s -H "X-aws-ec2-metadata-token: $TOKEN" http://169.254.169.254/latest/meta-data/public-ipv4)</h2>" >> /var/www/html/index.
echo "<h2>Public DNS: $(curl -s -H "X-aws-ec2-metadata-token: $TOKEN" http://169.254.169.254/latest/meta-data/public-hostname)</h2>" >> /var/www/html/i
echo "<h2>Deployed via Terraform</h2>" >> /var/www/html/index.html
```

```
# 3. Webserver Module Call - Nginx Proxy Server (Instance Suffix 0)
module "myapp-webserver" {
  source            = "./modules/webserver"
  env_prefix        = var.env_prefix
  instance_type     = var.instance_type
  availability_zone = var.availability_zone
  public_key        = var.public_key
  my_ip             = local.my_ip
  vpc_id            = aws_vpc.myapp_vpc.id
  subnet_id         = module.myapp-subnet.subnet.id
  script_path       = "./entry-script.sh"
  instance_suffix   = "0"
}

# 4. Webserver Module Call - Apache Backend Server (Instance Suffix 1)
module "myapp-web-1" {
  source            = "./modules/webserver"
  env_prefix        = var.env_prefix
  instance_type     = var.instance_type
  availability_zone = var.availability_zone
  public_key        = var.public_key
  my_ip             = local.my_ip
  vpc_id            = aws_vpc.myapp_vpc.id
  subnet_id         = module.myapp-subnet.subnet.id
  script_path       = "./apache.sh"
  instance_suffix   = "1"
}
```

```
output "webserver_public_ip" {
  value = module.myapp-webserver.aws_instance.public_ip
}

output "aws_web-1_public_ip" {
  value = module.myapp-web-1.aws_instance.public_ip
}
```

```
Apply complete! Resources: 3 added, 0 changed, 0 destroyed.

Outputs:

aws_web-1_public_ip = "158.252.32.32"
webserver_public_ip = "3.28.136.167"
```

```
The authenticity of host '3.28.136.167 (3.28.136.167)' can't be established.
ED25519 key fingerprint is [redacted]E.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '3.28.136.167' (ED25519) to the list of known hosts.

A newer release of "Amazon Linux" is available.
  Version 2023.10.20260105:
Run "/usr/bin/dnf check-release-update" for full release and version update info
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
```

```
user nginx;
worker_processes auto;
error_log /var/log/nginx/error.log notice;
pid /run/nginx.pid;

events {
    worker_connections 1024;
}

http {
    include /etc/nginx/mime.types;
    default_type application/octet-stream;

    server {
        listen 443 ssl;
        server_name 3.28.136.167;

        ssl_certificate /etc/ssl/certs/selfsigned.crt;
        ssl_certificate_key /etc/ssl/private/selfsigned.key;

        # --- UPDATED SECTION START ---
        location / {
            proxy_pass http://158.252.32.32:80;
            proxy_set_header Host $host;
            proxy_set_header X-Real-IP $remote_addr;
            proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
            proxy_set_header X-Forwarded-Proto $scheme;
        }
        # --- UPDATED SECTION END ---
    }

    server {
        listen 80;
        server_name _;
        return 301 https://$host$request_uri;
    }
```

```
[ec2-user@ip-10-0-10-51 ~]$ sudo systemctl restart nginx
[ec2-user@ip-10-0-10-51 ~]$
```

```
[ec2-user@ip-10-0-10-51 ~]$ cat /var/log/nginx/error.log
2026/01/17 19:24:43 [notice] 3012#3012: using the "epoll" event method
2026/01/17 19:24:43 [notice] 3012#3012: nginx/1.28.0
2026/01/17 19:24:43 [notice] 3012#3012: OS: Linux 6.1.158-180.294.amzn2023.x86_64
2026/01/17 19:24:43 [notice] 3012#3012: getrlimit(RLIMIT_NOFILE): 65535:65535
2026/01/17 19:24:43 [notice] 3055#3055: start worker processes
2026/01/17 19:24:43 [notice] 3055#3055: start worker process 3057
2026/01/17 19:24:43 [notice] 3055#3055: start worker process 3058
2026/01/17 19:24:44 [notice] 3055#3055: signal 3 (SIGQUIT) received from 1, shutting down
2026/01/17 19:24:44 [notice] 3057#3057: gracefully shutting down
2026/01/17 19:24:44 [notice] 3057#3057: exiting
2026/01/17 19:24:44 [notice] 3057#3057: exit
2026/01/17 19:24:44 [notice] 3058#3058: gracefully shutting down
2026/01/17 19:24:44 [notice] 3058#3058: exiting
2026/01/17 19:24:44 [notice] 3058#3058: exit
2026/01/17 19:24:44 [notice] 3055#3055: signal 17 (SIGCHLD) received from 3057
2026/01/17 19:24:44 [notice] 3055#3055: worker process 3057 exited with code 0
2026/01/17 19:24:44 [notice] 3055#3055: worker process 3058 exited with code 0
2026/01/17 19:24:44 [notice] 3055#3055: exit
2026/01/17 19:24:44 [warn] 3070#3070: could not build optimal types_hash, you should increase either types_hash_max_size: 1024 or types_hash_bucket_size: 64; ignoring types_h
ash_bucket_size
2026/01/17 19:24:44 [warn] 3906#3906: could not build optimal types_hash, you should increase either types_hash_max_size: 1024 or types_hash_bucket_size: 64; ignoring types_h
ash_bucket_size
2026/01/17 19:24:44 [notice] 3906#3906: using the "epoll" event method
```

Welcome to My Web Server

Hostname: myapp-webserver

Public IP: 158.252.32.32

Public DNS:

Deployed via Terraform

Task 8

```
# 5. Webserver Module Call - Second Apache Backend (Instance Suffix 2)
module "myapp-web-2" {
  source            = "./modules/webserver"
  env_prefix        = var.env_prefix
  instance_type     = var.instance_type
  availability_zone = var.availability_zone
  public_key        = var.public_key
  my_ip             = local.my_ip
  vpc_id            = aws_vpc.myapp_vpc.id
  subnet_id         = module.myapp-subnet.subnet.id
  script_path       = "./apache.sh"
  instance_suffix   = "2"
}
```

outputs.tf

```
output "webserver_public_ip" {
  value = module.myapp-webserver.aws_instance.public_ip
}

output "aws_web-1_public_ip" {
  value = module.myapp-web-1.aws_instance.public_ip
}
output "aws_web-2_public_ip" {
  value = module.myapp-web-2.aws_instance.public_ip
}
```

```
Apply complete! Resources: 3 added, 0 changed, 0 destroyed.

Outputs:

aws_web-1_public_ip = "158.252.32.32"
aws_web-2_public_ip = "3.28.44.105"
webserver_public_ip = "3.28.136.167"
```

Welcome to My Web Server

Hostname: myapp-webserver

Public IP: 3.28.44.105

Public DNS:

Deployed via Terraform

Welcome to My Web Server

Hostname: myapp-webserver

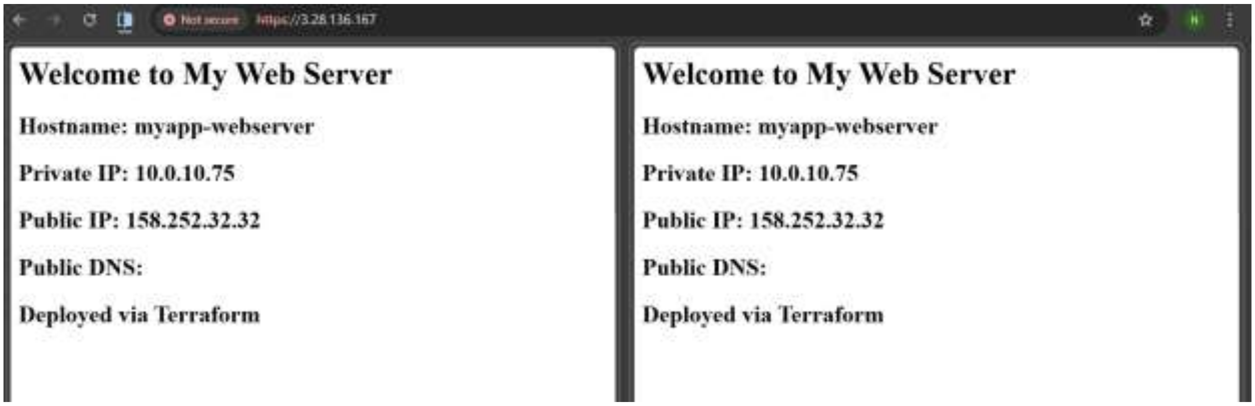
Public IP: 158.252.32.32

Public DNS:

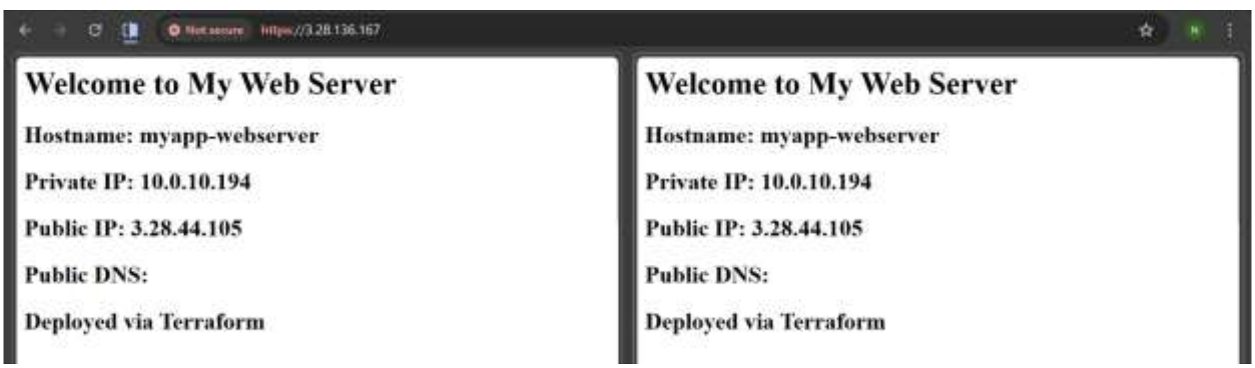
Deployed via Terraform

Task 9

```
# Define the group with Web-2 as BACKUP
upstream backend_servers {
    server 158.252.32.32:80;          # web-1 (Primary)
    server 3.28.44.105:80 backup;     # web-2 (Backup)
}
```




```
# Define the group with Web-1 as BACKUP
upstream backend_servers {
    server 158.252.32.32:80 backup;         # web-1 (backup)
    server 3.28.44.105:80 ;     # web-2 (Primary)
}
```



Task 10

```
location / {
    proxy_pass http://backend_servers;

    # 2. Enable caching for this location
    proxy_cache my_cache;
    proxy_cache_valid 200 60m;
    proxy_cache_key "$scheme$request_uri";

    # 3. Add header to see if it's a HIT or MISS in the browser
    add_header X-Cache-Status $upstream_cache_status;

    proxy_set_header Host $host;
    proxy_set_header X-Real-IP $remote_addr;
}
```

```
[ec2-user@ip-10-0-10-51 ~]$ sudo ls -la /var/cache/nginx/
total 0
drwx------. 2 nginx root   6 Jan 17 20:04 .
drwxr-xr-x. 9 root  root 101 Jan 17 20:04 ..
[ec2-user@ip-10-0-10-51 ~]$
```

Clean up :

```
module.myapp-subnet.aws_subnet.myapp_subnet_1: Destruction complete after 1s
module.myapp-web-2.aws_security_group.web_sg: Destruction complete after 1s
aws_vpc.myapp_vpc: Destroying... [id=vpc-07f83cc4f9b96672f]
aws_vpc.myapp_vpc: Destruction complete after 1s

Destroy complete! Resources: 13 destroyed.
```

```
{
  "version": 4,
  "terraform_version": "1.14.3",
  "serial": 90,
  "lineage": "2702fe6f-53f9-f4fd-4414-2e9d6d5a13fd",
  "outputs": {},
  "resources": [],
  "check_results": null
}
```

```
total 65420
drwxrwxrwx+ 6 codespace root          4096 Jan 17 20:14 .
drwxr-xrwx+ 5 codespace root          4096 Jan 17 18:11 ..
drwxrwxrwx+ 8 codespace root          4096 Jan 17 18:22 .git
drwxr-xr-x+ 4 codespace codespace     4096 Jan 17 19:10 .terraform
-rw-r--r--  1 codespace codespace     2422 Jan 17 18:35 .terraform.lock.hcl
-rw-rw-rw-  1 codespace root            11 Jan 17 18:22 README.md
-rw-rw-rw-  1 codespace codespace      926 Jan 17 19:28 apache.sh
drwxr-xr-x+ 3 codespace codespace     4096 Jan 16 19:26 aws
-rw-rw-rw-  1 codespace codespace 66875639 Jan 17 18:37 awscliv2.zip
-rw-rw-r--  1 codespace codespace     1705 Jan 17 19:23 entry-script.sh
-rw-rw-r--  1 codespace codespace      122 Jan 17 18:26 locals.tf
-rw-rw-r--  1 codespace codespace     2010 Jan 17 19:47 main.tf
drwxrwxrwx+ 4 codespace codespace     4096 Jan 17 19:13 modules
-rw-rw-r--  1 codespace codespace      259 Jan 17 19:47 outputs.tf
-rw-rw-rw-  1 codespace codespace      182 Jan 17 20:14 terraform.tfstate
-rw-rw-rw-  1 codespace codespace    41823 Jan 17 20:13 terraform.tfstate.backup
-rw-rw-r--  1 codespace codespace      219 Jan 17 18:27 terraform.tfvars
-rw-rw-r--  1 codespace codespace      196 Jan 17 18:26 variables.tf
```